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1. Introduction

1.1 Methodology

This project evaluates Canada's standing in global AI competitiveness using a data-driven benchmarking framework. Using publicly available datasets, expert reports, and standardized international indicators, we constructed comparable quantitative scores for each dimension. The benchmarking process involved data normalization, cross-country standardization, and the use of weighted composite indices to make results methodologically consistent across nations. This allowed us to compare Canada not only to global AI leaders such as the United States, China, and the EU, but also to emerging peer economies. We created quantitative scores across five areas: Research, Talent, Compute, Industry, and Policy. This allows us to compare Canada with leading nations. These metrics highlight specific strengths and structural gaps, providing a data-driven basis for strategic recommendations.

1.2 Findings & Analysis

Country	Research	Talent	Compute	Industry	Policy	Raw_Total	Softmax_Score
United States	0.771252	1.0	0.959928	0.773879	0.926705	0.889886	0.202568
China	0.61802	0.194811	0.101299	0.272261	0.091304	0.278002	0.109858
United Kingdom	0.256311	0.180254	0.057407	0.301807	0.573913	0.251981	0.107036
France	0.164031	0.115731	0.07079	0.304421	0.368336	0.185012	0.100103
South Korea	0.272766	0.077766	0.049113	0.268173	0.288402	0.180942	0.099696
Germany	0.246985	0.131884	0.190429	0.095315	0.100253	0.162138	0.097839
Canada	0.23006	0.149036	0.032381	0.100389	0.175142	0.14258	0.095944
Japan	0.094831	0.037944	0.188736	0.068388	0.308455	0.127467	0.094505
India	0.15	0.18352	0.016739	0.045339	0.079687	0.105482	0.09245

Figure 1. Table: Multidimensional AI Capability Scores by Country

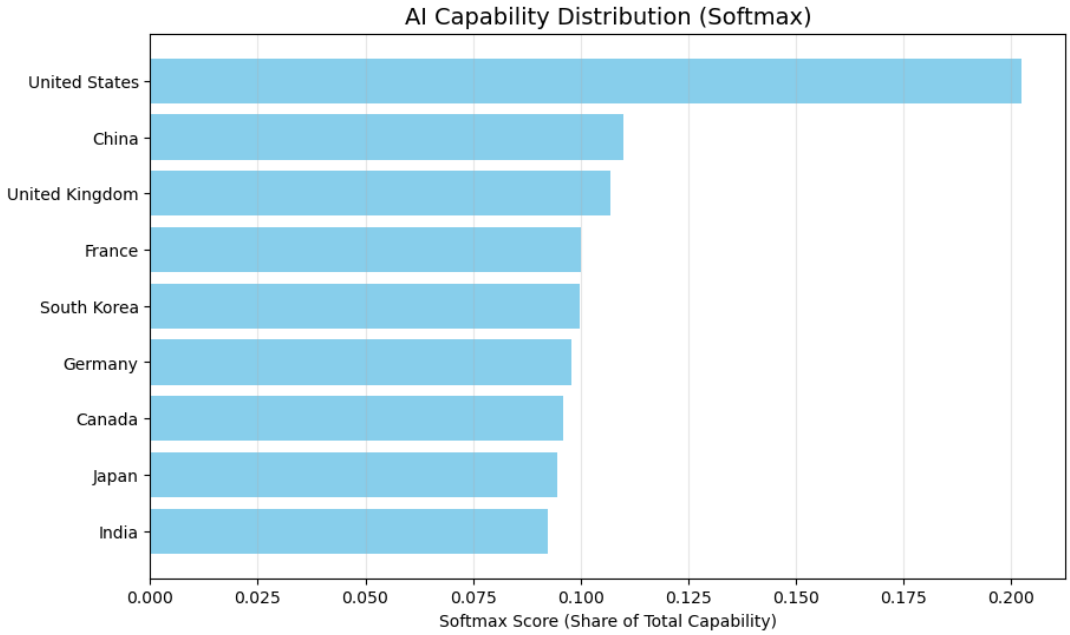

Figure 2. AI Capability Distribution (Softmax) by Country

The Research score evaluates innovation capacity and academic impact using publication and citation indicators. We included each country's share of the global top-100 most-cited papers, the 2021–2023 average publication volume, and per-capita publication intensity. We also measured long-term publication growth. The final Research Score shows a clear U.S. lead, strong Chinese growth, and a mid-tier but stable Canadian performance.

Our analysis of the Talent ecosystem relies on 12 datasets that cover workforce strength and concentration. By creating a weighted competitiveness index, we found that Canada performs moderately overall, with healthy hiring and job posting activity. However, significant weaknesses emerge when we look at the broader ecosystem. Canada falls behind the U.S., China, and the U.K. in robotics density, investment scale, and startup formation.

Infrastructure proves to be a significant bottleneck. Using the global TOP500 supercomputing database to measure computational capacity, we found a high concentration of power in the United States, followed by Germany and Japan. Canada's presence is small, mostly limited to academic and government sectors. This marks one of the most significant competitive gaps in our analysis.

The Industry score measures the strength of commercialization based on venture capital investment in AI from 2015 to 2025. We calculated two indicators: the average VC investment level and the percentage growth. The final Industry Score shows that the U.S. is the clear global leader, with strong secondary clusters in France, the U.K., China, and Korea. Canada ranks low because of smaller funding amounts and slower commercialization growth, highlighting limited translation of research into market activity.

Finally, regarding national strategy, we tracked legislative involvement by counting AI-related bills and official mentions. While the U.S. and U.K. show strong policy movement, Canada has moderate visibility and lower concrete legislative activity. Sectoral data indicates that while Canada invests in health, education, and defense, these investments are modest, highlighting the need for a more coordinated national approach.

2. Canada Innovation Strategies

Canada has launched a national level AI strategy since 2017, which is one of the earliest in the world. The current public service AI strategy proposed by the federal government for 2025-2027 clearly emphasizes two directions: "responsible adoption of AI" and "expanding Canada's AI talent and capabilities". In this context, we analyzed and generated relevant word clouds. The heavy use of words such as' technology ',' compute ',' model ', and' infrastructure 'in the word cloud reflects Canada's emphasis on' technology and infrastructure capabilities' as a long-term strategic priority. At the same time, the words "talent," "skills," "training," "implementation," and "deployment" in the word cloud highlight the need for Canada to address research agility and talent bottlenecks in the short term. Multiple governance studies have pointed out that although Canada is leading in AI research and talent development, there are still significant gaps in "adopting scale," "deploying and transforming," and "building digital infrastructure. Overall, the weight structure of word cloud presents a "dual track" strategic logic: Releasing competitiveness through "talent, deployment, and training" in the short term; Build sustainable advantages in the medium and long term through "technological infrastructure, model capabilities, and governance structures".

Figure 3. Word cloud generated based on government reports

Also combined with our analysis from Part 1 which shows that Canada’s core challenges fall into two categories: short-term bottlenecks that limit Canada’s competitiveness today, and long-term capacity gaps that will determine our position over the next decade. Based on this distinction, we divide our national AI capability enhancement plan into short-term measures that focus on improving research agility and expanding AI talent capacity, and long-term initiatives that target the foundational enablers of AI competitiveness, particularly compute infrastructure and industry-wide adoption. Therefore, we conducted strategy analysis and strategy formulation for the following two periods.

2.1 Short-Term Strategy for Sustained AI Competitiveness

Strengthening Canada’s AI research capacity is a core short-term priority, especially given its weak comparative performance in our Part 1 research index. Using a combined research-score reflecting

publication output, global visibility, and innovation potential, Canada ranked 7th out of 9 major AI nations with a score of 0.0247—far behind China (0.7783) and the United States (0.7185). Despite hosting world-class institutes such as Mila, Vector, and Amii, Canada's overall research strength remains modest on the international stage. Its impact score is only 0.036, supported by just 6 highly cited papers among the global top 100—far below the United States (50) and China (34). Canada also records the lowest publication volume score (0.000) across all countries in our dataset, and its growth score (0.0199) remains marginal compared to China (0.799) and India (1.000). These indicators reflect limited scale, slow research expansion, and weak global visibility despite strong academic institutions. These limitations translate into clear structural gaps that Canada must address in the short term. The absence of scale and low publication volume highlight the need for more agile, rapid-turnaround research funding to accelerate output. Canada's modest applied research footprint and limited growth trajectory point to a need for expanded applied AI research capacity, particularly through programs that can generate pilot projects and sector-specific innovation. Finally, the disconnect between Canada's respected institutes and its low global impact score underscores the urgency of strengthening pathways that translate academic work into industry and public-sector applications.

However, its per-capita performance reveals a more nuanced picture. When adjusting for population size, Canada achieves a per-researcher score of 0.2301, comparable to Germany (0.2470) and the United Kingdom (0.2563), and well above Japan (0.0948) and India (0.1500). This suggests that Canada's core researchers are productive and competitive, but the surrounding research ecosystem lacks the scale and structural support needed to amplify their impact.

Importantly, the structural weaknesses observed in the research system mirror the constraints evident in Canada's AI talent ecosystem. Part 1 shows that Canada holds a stable but uncompetitive position in global AI talent. Its talent concentration (0.93) and skill penetration (1.41) indicate a capable technical workforce and strong academic foundations. However, Canada's representation score (32.22) is far below leading countries such as the U.S. (33.68) and particularly the U.K. (70.50), reflecting limited global influence and lower visibility among AI researchers. Despite strong labor demand — evidenced by high job posting intensity (1.41) and an exceptionally strong hiring rate (26.13) — Canada's migration score (0.95) trails Germany (2.13) and the U.S. (1.07). This imbalance shows that Canadian firms urgently need AI specialists, yet Canada is not a preferred destination for globally mobile talent.

Ecosystem metrics further reinforce these constraints. Canada's annual AI investment (2.887) and cumulative stock (15.3) lag dramatically behind the United States (109 and 470). Startup formation is low (51 vs. 1073 in the U.S.), limiting opportunities for researchers and practitioners and reducing Canada's talent absorption capacity. These factors anchor Canada's PCA Talent Score at 0.149, significantly behind the U.S., China (0.194), and India (0.184). To stay competitive, Canada must quickly address its short-term weaknesses in attraction, retention, and applied talent capacity — the most sensitive and actionable elements of the talent ecosystem.

2.2 Long-Term Strategy for Sustained AI Competitiveness

While the short-term strategy focuses on immediate challenges in research agility, applied talent, and early-stage adoption, Canada's long-term competitiveness will ultimately depend on addressing deeper structural gaps identified in Part 1. Our benchmarking across major AI economies reveals persistent disadvantages in compute capacity, industrial scale, and commercialization readiness.

Canada ranks near the bottom among peer nations in **compute competitiveness**, with a composite score far below (0.05) not only the United States—whose score approaches 0.95—but also below mid-tier economies such as France, the U.K., and South Korea. This imbalance underscores the structural

constraints limiting Canada's ability to build, deploy, and scale advanced AI systems. The guiding goal is to establish sovereign, high-performance compute capacity that supports large-scale model training, sectoral deployment, and long-term digital autonomy. This requires building national compute infrastructure, ensuring interoperability and workload localization, and creating a data-and-model ecosystem that enables widespread adoption across industry, government, and research.

On the industry side, Canada's venture activity also trails leading nations. Part 1 results show Canada placed in the lower tier of the **AI industry competitiveness ranking**, with a VC-driven score around **0.10**, far below leaders such as the United States (~0.77), France (~0.30), and the U.K. (~0.30). Moreover, in the VC "level vs. growth" positioning chart, Canada appears as a low-level, modest-growth ecosystem, indicating that without targeted long-term interventions, domestic companies may struggle to reach scale. Given Canada's weak VC competitiveness and limited international growth, the strategy focuses on cultivating global AI champions, improving commercialization and export readiness, and supporting international market expansion. Over the next decade, Canada aims to build a stronger industrial base, increase AI adoption across high-impact sectors, and grow firms capable of competing in major markets such as the U.S., EU, and Asia.

3. Canada Innovation Implementations and Policies

3.1 Short-Term: Research Implementation

To implement Canada's short-term AI research strategy in practice, we propose a set of concrete, Canada-focused measures that can be introduced within 1–3 years. These steps are designed to increase research agility, support applied AI projects, and strengthen research–industry collaboration.

Figure 4. Canada AI research implementation plan

3.1.1 Research Responsiveness and Global Integration

We recommend establishing a Fast-Track AI Research Fund to support small, time-sensitive AI research projects with significantly shorter review cycles (e.g., 45–60 days). This mechanism would prioritize emerging areas such as generative AI, AI safety, healthcare AI, and AI for public services, enabling researchers to respond quickly to new opportunities without being constrained by traditional,

slow funding mechanisms. In parallel, Canada should strengthen international and diaspora research links by supporting short-term joint calls for AI research, visiting fellowships, and collaborations with Canadian researchers abroad. These combined measures ensure that Canadian researchers can stay aligned with global frontier innovation while operating within a faster, more flexible domestic funding environment.

3.1.2 Applied AI Scale-Up Mechanisms

To expand applied AI capacity and deepen industry–academic collaboration, Canada should modernize Mitacs and invest in sector-specific collaboration hubs. Building on Mitacs' proven track record—over 99,000 internships, 35,000+ innovation projects, and $1.42B in R&D investment between 2018 and 2025 (Mitacs Strategic Plan 2026–2030)—the program can be positioned as a central platform for short-term applied AI research. Enhancements could include an AI-focused research stream, targeted support for projects requiring compute resources, and expanded collaboration with SMEs and public-sector organizations seeking AI solutions. Complementing these upgrades, we recommend establishing Applied AI Collaboration Hubs in key sectors such as healthcare, finance, manufacturing, and climate. Each hub would coordinate joint pilot projects between universities, companies, and government agencies, providing shared access to data, compute, and technical expertise while enabling 6–12 month testing of AI solutions in real-world environments.

3.1.3 Research-to-product Translation

Finally, Canada should strengthen the pathway from research discovery to real-world deployment. To reduce the persistent gap between academic research and application, we propose introducing a structured Research-to-Product Pathway that funds early-stage validation, user testing, regulatory preparation, and pilot implementation of AI solutions. Funding would be directed toward projects with clear end users in industry or the public sector, with a focus on moving prototypes toward actual adoption and commercialization outcomes. This downstream pathway ensures that upstream investments—in fast-track funding, Mitacs-based collaboration, and applied AI hubs—ultimately translate into tangible societal and economic benefits.

3.2 Short-Term: Talent Implementation

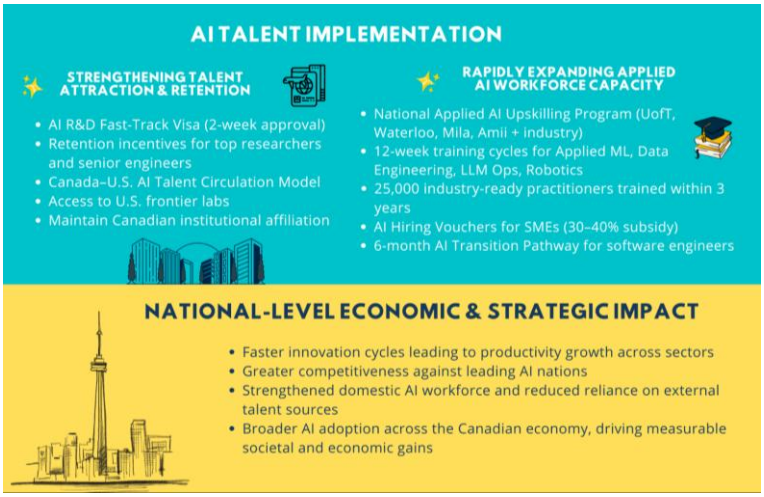

Figure 5. Canada AI Talent implementation plan

3.2.1 Strengthening Attraction and Retention

Given low migration and representation scores, improving Canada's attractiveness to top AI talent is a clear priority. While Canada cannot match U.S. compensation or ecosystem scale, it can enhance its value proposition through targeted interventions.

Stabilizing domestic high-end talent is essential. Retention incentives can offset pay gaps and help Canada retain core researchers and engineers. Likewise, a fast-track AI R&D visa with a two-week approval cycle would significantly reduce barriers for international talent and move Canada closer to competitiveness with the U.K. and Singapore.

Canada's proximity to the U.S. also presents an opportunity. Instead of treating the U.S. as a talent drain, Canada can create a Canada–U.S. AI talent circulation model, enabling Canadian researchers to participate in U.S.-based frontier projects while maintaining Canadian institutional ties. This approach transforms geographic disadvantage into strategic collaboration and strengthens Canada's knowledge base without requiring ecosystem scale equivalent to the U.S.

Together, these measures directly improve migration, representation, and PCA Talent Score components tied to attraction and global competitiveness.

3.2.2 Rapidly Expanding Applied AI Workforce Capacity

Canada's most acute bottleneck lies in the applied talent layer — the practitioners who deploy AI in real-world settings. Although penetration is moderate (1.41), the extremely high hiring rate (26.13) suggests severe workforce shortages. Without rapid expansion of applied AI talent, Canadian firms will continue to face implementation delays, undermining competitiveness across sectors.

To address this, Canada should establish a national applied AI upskilling program delivered through partnerships with UofT, Waterloo, Mila, and industry. These 12-week cycles would train workers in applied machine learning, data engineering, LLM operations, and robotics integration, aiming to produce 25,000 industry-ready practitioners within three years. This would meaningfully raise penetration scores and ease hiring pressures.

Given weak ecosystem indicators — low investment and limited startup activity — Canada must also strengthen enterprise absorption capacity. AI Hiring Vouchers for SMEs can stimulate immediate demand by helping companies fund AI talent and adoption projects. A six-month AI transition track for software engineers further expands supply at low cost and short training duration. These initiatives reinforce penetration, hiring efficiency, and ecosystem activity — key contributors to Canada's PCA talent improvement.

3.3 Long-Term: Industry Implementation

Canada's long-term AI industry strategy requires a coordinated plan that connects regional clusters, empowers SMEs, and supports global expansion. The overarching goal is to double AI employment from about 140,000 to 300,000–400,000 by 2035 and achieve 45–60% AI adoption across key sectors.

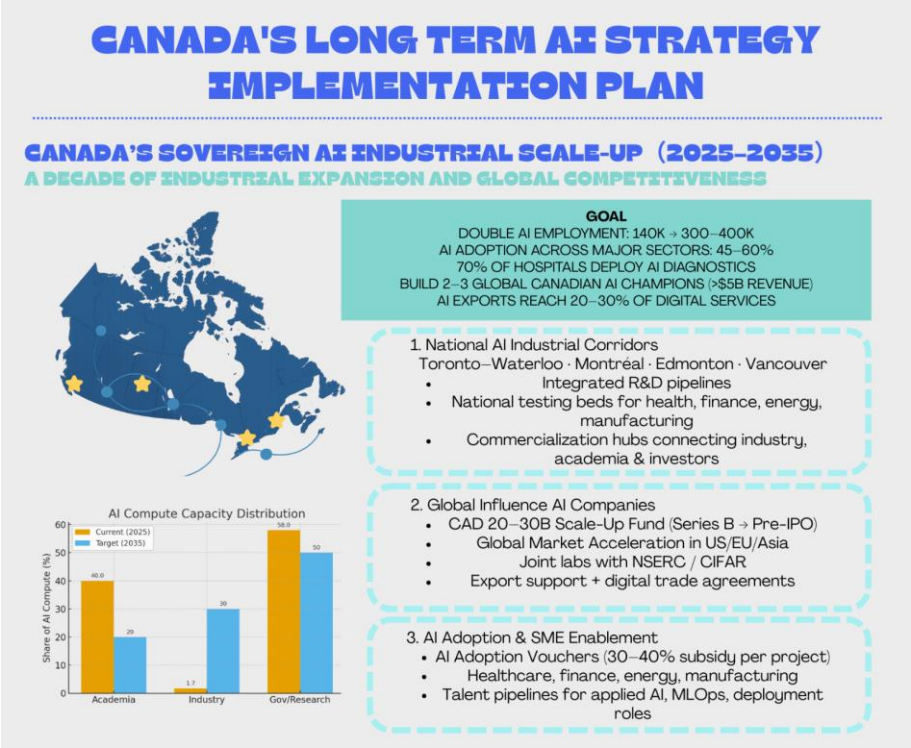

Figure 6. Canada AI industry implementation plan

3.3.1 Establishing National AI Industrial Corridors

The first step is to formally establish Canada's National AI Industrial Corridors, which will consolidate existing clusters such as the Toronto–Waterloo Corridor, Montréal AI Hub, and Alberta Machine Intelligence Institute (Amii) into a coordinated national network. Led jointly by the Ministry of Innovation, Science and Economic Development (ISED) and provincial governments, this initiative seeks to integrate universities, incubators, investors, and private enterprises into cohesive ecosystems. Each corridor will develop applied research and testing facilities, fostering cross-sector collaboration in healthcare, finance, energy, and manufacturing. This structure mirrors successful international models such as the EU's Digital Innovation Hubs and the U.S. Regional Technology Hubs, which have effectively linked research excellence with commercial innovation. By coordinating R&D, infrastructure, and policy frameworks, these corridors will strengthen Canada's industrial foundation and accelerate sector-wide AI adoption.

3.3.2 Promoting SME Adoption and Workforce Growth

To ensure AI adoption extends beyond large corporations, the federal government should implement an AI Adoption Voucher Program for small and medium-sized enterprises (SMEs). Administered by the Business Development Bank of Canada, this program would subsidize up to 30–40% of AI-related costs—covering training, model deployment, and compute usage—up to CAD 200,000 per firm. Drawing lessons from Singapore and South Korea, such targeted incentives have proven effective in accelerating SME-level digital transformation. Lowering financial barriers will encourage experimentation, stimulate regional innovation, and expand employment in AI-related roles such as data engineering and model operations. Over time, these measures will directly contribute to the national goal of doubling AI workforce participation by 2035.

3.3.3 Expanding Canadian AI companies with global influence

The final component focuses on scaling up high-potential Canadian AI companies and bridging the commercialization gap. A Canada AI Scale-Up Fund of CAD 20–30 billion should be established to support firms in the Series B to Pre-IPO stages, drawing capital from both public (CPP, OTPP) and private investors. This fund would prioritize high-impact fields such as healthcare AI, energy optimization, and enterprise software. In parallel, a Global Market Acceleration Program, managed by Global Affairs Canada, should create international innovation hubs in Boston, Berlin, and Singapore to help Canadian firms access foreign markets, offering regulatory guidance, brand support, and partial marketing subsidies. Furthermore, collaboration between CIFAR, NSERC, and scaling firms through joint research programs and tax incentives will ensure continued innovation. Together, these measures will enable Canada to build two to three globally competitive AI enterprises and increase AI exports to 20–30% of total digital services by 2035.

3.4 Long-Term: Compute Implementation

Beyond industrial growth, long-term competitiveness depends on sovereign compute capacity and secure data infrastructure. Canada must expand national compute resources, localize sensitive workloads, and build public AI assets accessible to all sectors.

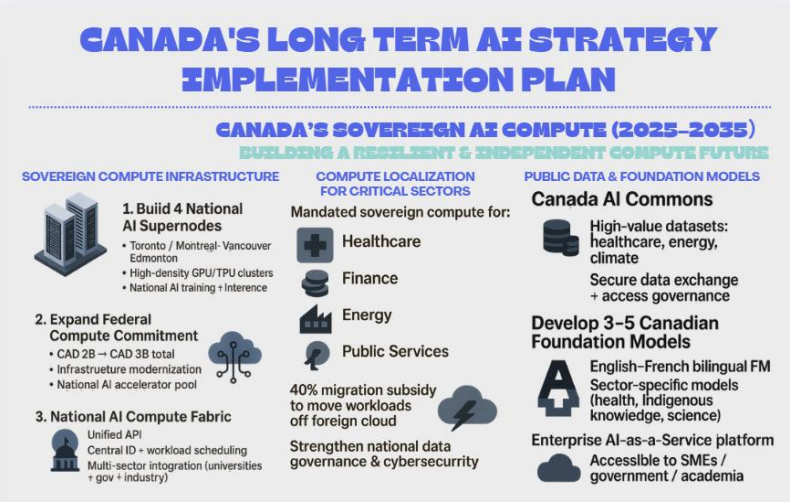

Figure 7. Canada AI compute implementation plan

3.4.1 Building Sovereign Compute Infrastructure

The first priority is to construct National AI Supernodes in Toronto, Montréal, Vancouver, and Edmonton, forming a distributed yet interconnected national compute network. Managed by ISED, the Canada Innovation Corporation, and Compute Canada, these supernodes will host advanced GPU/TPU clusters, high-bandwidth interconnects, and energy-efficient cooling systems. The federal government's CAD 2 billion sovereign compute commitment should be expanded by an additional CAD 1 billion to ensure regular hardware and network upgrades.

To connect these resources, Canada should establish a National AI Compute Fabric, integrating compute assets from universities, research institutions, and government agencies. A unified API, centralized authentication, and elastic workload scheduling would enable efficient sharing and equitable allocation of compute capacity nationwide. Long-term sustainability would be ensured through a Sovereign Compute Infrastructure Fund, managed by the Department of Finance and the Canada Infrastructure Bank, providing stable capital for ongoing upgrades under a public–private partnership model.

3.4.2 Localizing Compute for Critical Sectors

A Critical Sector Compute Localization Policy should mandate that AI training and inference in healthcare, finance, energy, and public services be hosted within sovereign infrastructure. Migration subsidies covering up to 40% of transition costs and temporary fee discounts would encourage firms to shift from foreign cloud platforms, improving data security and regulatory compliance.

3.4.3 Developing Public Data and Foundation Models

The Canada AI Commons, managed by Statistics Canada and ISED, should integrate high-value datasets from healthcare, energy, and climate sectors to support national foundation model training. Using sovereign compute, three to five Canadian foundation models—including multilingual and sector-specific versions—should be developed and made accessible via an Enterprise AI-as-a-Service platform. This will democratize AI adoption, reduce costs for SMEs, and ensure equitable access to advanced AI capabilities.

4. Conclusion and Expected Impact

This integrated strategy is expected to deliver a step-change in Canada's AI competitiveness. In the short term, faster research cycles, expanded applied AI capacity, and improved talent attraction will directly increase Canada's innovation velocity, strengthen its technical workforce, and unlock AI adoption across industry and the public sector. These changes are projected to raise Canada's PCA Talent and Research scores, increase migration inflows, and accelerate the number of viable AI pilots and deployments nationwide. In the long term, investments in sovereign compute, commercialization pathways, and industry scale-up will build a durable foundation for national AI leadership—expanding Canada's domestic AI production capacity, enabling globally competitive firms to emerge, and reducing structural reliance on foreign compute and technologies. Ultimately, the combined impact of these actions positions Canada to transition from a mid-tier AI participant into a high-performing, innovation-driven AI economy capable of sustaining productivity growth, attracting global capital, and competing with leading AI nations.